

Creativity-Expansive Tooling: Design-Led Approaches to Algorithmic Design for Fabrication

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Figure 1: This dissertation develops fabrication tools whose algorithms and representations engage actively in making—proposing configurations, reframing structural reasoning, and producing outcomes through negotiation between computational behavior and material practice: (a) object-driven shelving layouts generated around personal artifacts; (b) shelving designed to negotiate between the environment, plants, and a cat; (c) algorithmically curved shelving realized with bent lamination; and (d) generated voids realized as ceramic objects.

Abstract

Making has always involved negotiation—with tools, physical materials, and emergent constraints. Yet tool designers typically position computation as a means of execution: the practitioner specifies, the tool produces. My dissertation develops *creativity-expansive tooling*: fabrication tools whose algorithms and representations propose, constrain, and reframe—acting as interlocutors in a negotiation between practitioner intent, computational behavior, and material response. I design and evaluate two systems: *Uniquely Shaped Spaces*, a generative shelving tool whose algorithm proposes configurations shaped by its own tendencies, analyzed through seamful design; and *Cross-Section Weaving*, a yarn-centric textile design tool whose notation reframes how weavers reason about structure. Across both, I analyze *moments of negotiation*—instances where a practitioner encounters computational logic or material response and acts to redirect the workflow—examined through longitudinal study, artifact analysis, and autoethnography, to develop design knowledge for researchers and designers of computational fabrication tools.

CCS Concepts

• **Human-centered computing** → **Interactive systems and tools.**

Keywords

Computational Design, Digital Fabrication, Hybrid Craft, Computational Systems

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1 Introduction

As computation increasingly mediates creative practice—from algorithmic design [4, 8, 19] to machine-assisted making [3, 17, 20]—human-computer interaction (HCI) faces a recurring question: how can computational tools expand what practitioners can imagine while preserving the judgment and material engagement that make craft meaningful? But computation is not a neutral medium. When algorithms generate forms, constrain structures, and mediate between intention and material, they introduce their own logic into making—and practitioners must navigate a tool that is not just an instrument but an additional material to be negotiated. Researchers and tool designers building fabrication systems often position computation as a means of execution: the practitioner specifies, the



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tool produces, and the relationship flows in one direction through heterogeneous CAD/CAM toolchains [22–24]. Yet making with physical materials has never been pure execution—practitioners negotiate with resistance, surprise, and emergent possibility throughout [12]. When computational tools treat this negotiation as noise to be eliminated rather than signal to be amplified, they narrow not just what can be revised, but what can be discovered through the act of making.

Building on Andersen et al.’s notion of “digital crafts-machine-ship”—creative collaboration that shares autonomy between crafts-person and machine [1]—I develop a thesis about *creativity-expansive tooling* in computational craft systems design. Guided by practices of HCI and research through design, I design and evaluate fabrication systems as interlocutors in making—algorithms and representations with their own logic that propose, constrain, and reframe—to understand how making unfolds as a negotiation between practitioner intent, computational behavior, and material response, and what becomes possible when tools participate actively in making rather than executing passively. In these systems, practitioners discover what they want through encounter with the tool’s computational logic, rather than specifying intent fully before execution [16]. I develop this understanding through systems as research artifacts and reflecting on what they make possible in practice [10], alongside empirical studies and technical evaluations grounded in concrete implementation details [13]. I call tools that establish this relationship “creativity-expansive”: they expand what is thinkable and makeable not by offering more options within a fixed design space, but by making computation and material into interlocutors whose behavior reshapes the practitioner’s goals midstream.

I operationalize this thesis through two systems, each embodying a distinct form of computational engagement in making. *Uniquely Shaped Spaces* [11] uses an object-driven generative pipeline to propose fabrication-valid shelving configurations that carve around personal artifacts. The algorithm is not a neutral design tool—it has its own tendencies, generating configurations shaped by void-driven geometry and stochastic optimization that the practitioner negotiates with as objects, constraints, and fabrication realities shift. I frame this system through seamful design: rather than pursuing seamlessness that hides infrastructure and translation work, seamful design surfaces the boundaries where heterogeneous logics (e.g. maker intent, machine logic, and material agency) meet so they remain inspectable and changeable [5, 15]. I use seams to name these boundaries in the shelving pipeline—points where representations shift or decisions become committed—and design them as places where the algorithm’s proposals can be inspected and redirected.

Cross-Section Weaving introduces a yarn-centric representation for designing textile structures through cross-section notation. Users describe how yarn moves through the structure—its path over and under warp across layers—rather than specifying loom instructions directly. Where *Uniquely Shaped Spaces*’ algorithm proposes configurations the practitioner negotiates with (generative engagement), the cross-section notation reframes how weavers think about structure (representational engagement): the notation changes what structures are thinkable, and the weaver’s intent forms through composing in the notation, not before it. Compilation translates the cross-section into a loom draft, and weaving reveals how the structure actually behaves in material—a feedback

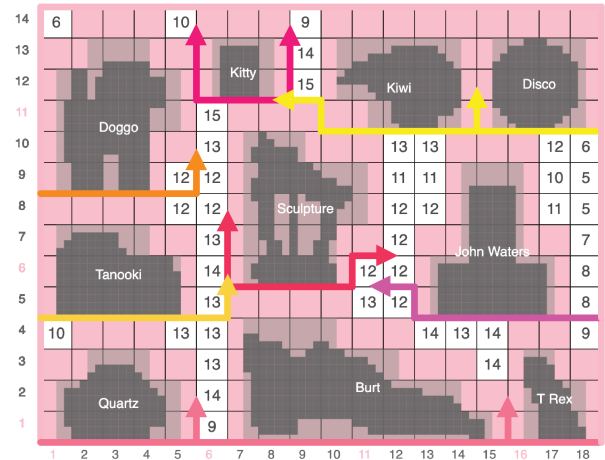


Figure 2: The algorithmic wall growth within the *Uniquely Shaped Spaces* interface.

loop where the practitioner negotiates between what the notation made thinkable and what the textile does.

Together, these two forms of computational engagement extend prior work on interactive fabrication interfaces and craft-forward workflows [21, 25] by designing computation not as a transparent executor of practitioner intent, but as an interlocutor with its own logic that shapes the negotiation.

Across both projects, I analyze what I call *moments of negotiation*: instances where a practitioner encounters the tool’s computational logic or the material’s response, interprets what it proposes or resists, and acts to redirect the workflow. This shared analytical lens—asking what prompted attention, how the practitioner interpreted the situation, and what action followed—connects findings across systems whose forms of computational engagement differ, and grounds the *creativity-expansive tooling* concept in observable practice. My near-term work focuses on completing a practitioner evaluation of *Cross-Section Weaving* and prototyping a third system that extends the *creativity-expansive tooling* thesis to a new material process and computational approach.

2 Uniquely Shaped Spaces

Uniquely Shaped Spaces is an object-driven system that generates custom, unusually-shaped shelving around users’ personal artifacts with walls that carve around each artifact (Fig. 2). The tool treats object silhouettes as direct inputs to generation: it translates their outlines into void constraints for a stochastic layout optimizer (simulated annealing) and a deterministic wall-growth procedure (cellular automata), allowing users to browse many possible shelving designs. Users move through the generative workflow by photographing and masking artifacts, generating shelving designs, curating objects and layout designs, and fabricating a selected design. The algorithm actively engages in this process: its stochastic optimization and void-driven growth produce configurations with their own tendencies—unexpected compartment shapes, densities, and spatial relationships—that the practitioner discovers and negotiates with rather than specifying in advance. This work extends

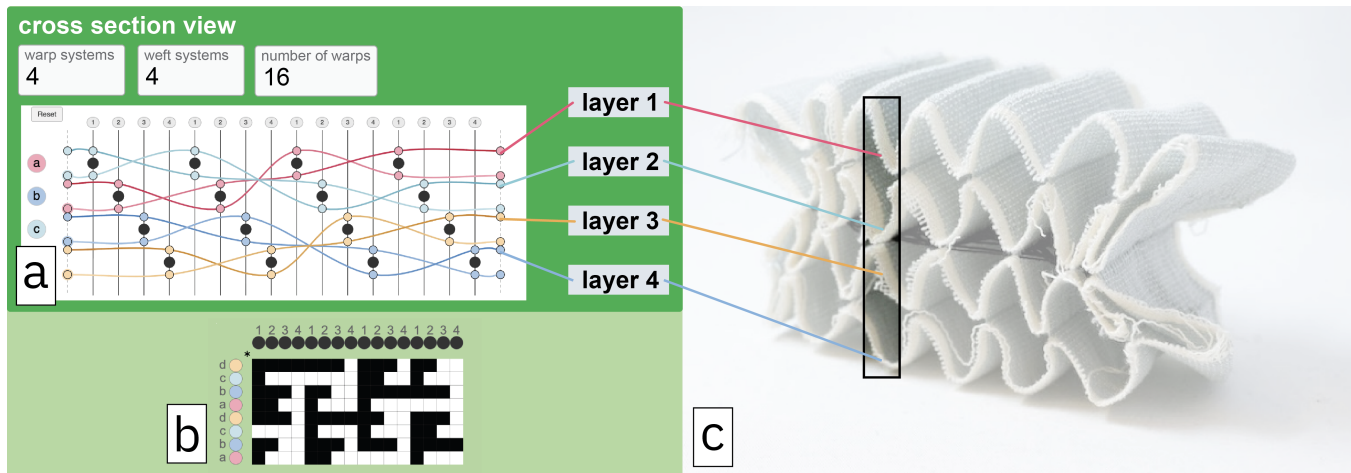


Figure 3: (a) A four-layer cross-section drawn in AdaCAD and (b) the corresponding weaving draft (lift plan) produced by compilation. (c) A four-layer elastic woven sample by weaver-artist Elizabeth Meiklejohn; the highlighted region indicates the portion defined by the AdaCAD cross-section.

human-in-the-loop fabrication research [6, 7] by demonstrating how object-driven constraints support storytelling and ongoing curation—and how practitioners push back on generativity by inspecting and reappropriating its logic. During a five-participant, twelve-week evaluation, we found that participants treated the algorithm’s proposals as prompts rather than solutions: they iteratively curated inputs to steer outcomes, discovering what they wanted through encounter with the algorithm’s behavior. For example, in Figure 1a, a participant inspected how void sizes shaped the generator and trimmed imported silhouettes to create smaller voids that better matched her goals—a goal she formed through seeing what the algorithm proposed, not before.

Through an autoethnographic follow-on inquiry (two shelves and a set of ceramic vessels), I found that “meaningful” generativity in fabrication is not solely output diversity but ongoing negotiation among maker intent, machine commitments, and material pushback. I made negotiation more explicit by designing seams organized around three axes in the pipeline. First, to foreground maker intent, I introduced a fixed exterior perimeter so the generator could align with higher-level user goals (e.g., window bounds or spatial constraints) rather than only packing artifacts. Second, to surface machine influence, I added a procedural algorithm that generates simple curved shelving forms, shifting the design space toward geometries that required negotiating fabrication through bent lamination. Third, to engage material pushback, I developed an export workflow that reinterprets generated voids as tangible objects, enabling clay 3D printing of modular nesting vessels. Across artifacts—a window-bounded plant shelf negotiated around a cat’s perch (Fig. 1b), a curved wooden shelf carried through planing, jointing, joinery, bent lamination, and CNC mold making (Fig. 1c), and clay 3D-printed vessels derived from void geometry (Fig. 1d)—these seams function as concrete interaction points where intent, algorithmic influence, and material realities can be inspected, redirected, and preserved as trace in both workflow documentation and the physical artifact.

Together, this work demonstrates one form of computational engagement in making: an algorithm that proposes across a generative pipeline, with seams designed as points where the practitioner can inspect and redirect its proposals—producing outcomes none of these actors could achieve alone. But this raises a question the shelving work alone cannot answer: what happens when the tool engages differently—not an algorithm that proposes configurations, but a representation that reframes how a practitioner thinks? *Cross-Section Weaving* investigates this complementary form of engagement.

3 Cross-Section Weaving

Cross-Section Weaving explores how computational representations can expand the structural vocabulary of woven textiles. Weave structure—the pattern of interlacement between warp and weft—determines everything from surface appearance to mechanical behavior, yet the dominant computational representation, the loom draft, describes what the machine should do rather than what the yarn will do [2, 18, 26]. Experienced weavers can read a draft and anticipate the resulting fabric, though for multilayer structures, this legibility becomes more challenging, requiring simultaneous mental tracking of which threads belong to which layer, where layers interchange, and how lift sequences map onto three-dimensional textile behavior. The draft faithfully describes what the loom will execute, but it does not represent what the weaver is trying to achieve.

Rather than designing multilayer structures through drafts, I developed a cross-sectional design modality within AdaCAD [8, 9] that lets practitioners specify structure directly and compile it into files for a computer-controlled loom that lifts individual warp threads (the vertical threads) (Fig. 3). The cross-section view is intentionally exact: it describes how the weft (the horizontal thread inserted row by row) travels over and under warp across layers, affording designs that would be very challenging to create from

traditional top-down drafting. Compilation of sketches then translates these structures into drafts that update live during drawing and advance the cloth row by row.

I call this a “yarn-centric representation”: a computational representation grounded in the yarn’s behavior rather than the loom’s instruction set. In early use by myself and collaborating weavers, the notation has proven to be not a passive recording medium but an active shaper of the design process, changing what structures the weaver can think about and propose. By composing layers, intersections, and binding points in cross-section, weavers reason about structures that would be very difficult to conceive through traditional top-down drafting; their intent forms through working in the notation, not before it. Where *Uniquely Shaped Spaces* embodies generative engagement—an algorithm that proposes configurations with its own tendencies—*Cross-Section Weaving* embodies representational engagement: a notation that reframes structural thinking and makes new categories of structure thinkable. Compilation then tests whether a proposed structure survives translation to loom instruction, and sampling—weaving a small textile—provides a material feedback layer, as the fabric’s behavior may confirm or contradict the compiled draft. There is no computational evaluation of whether a structure “works”; the material speaks at fabrication, not in the interface, making the weaver’s encounter with the woven result a distinctive form of negotiation. For example, in Figure 3a a weaver sketches a section of a four-layer cross section, compiles it into a draft (Fig. 3b), weaves a small swatch (Fig. 3c), and revises layer transitions in the cross-section based on the resulting textile’s behavior.

To study how this modality supports creative practice, I plan an evaluation combining three approaches. First, a hands-on workshop with experienced digital weavers who already work with Jacquard looms and multilayer designs, examining how they alternate between structural composition, compilation, and sampling. Second, a longitudinal follow-up with two to three workshop participants who continue working with the cross-section modality in their own studios, extending engagement beyond a single session to surface the steering strategies that sustained use reveals. Third, autoethnographic making that documents my own sustained use of the tool to develop multilayer structures. Process traces—cross-section drawings, compiled drafts at successive revisions, woven swatches, and revision notes—will serve as primary data for analyzing *moments of negotiation* between structural intent, compilation constraints, and material behavior.

4 Future Work

The remainder of the dissertation develops *creativity-expansive tooling* as intermediate-level design knowledge [14]: a strong concept for researchers and designers of computational fabrication tools that offers design patterns for structuring computation as an interlocutor in making, analytical frames (*moments of negotiation*) for studying how practitioners steer through algorithmic and material encounters, and empirical narratives of what becomes possible when tools participate actively in creative practice. The two systems presented here demonstrate distinct forms of computational engagement—generative engagement through an algorithm

that proposes across a pipeline in *Uniquely Shaped Spaces*, representational engagement through a notation that reframes structural thinking in *Cross-Section Weaving*—and the dissertation asks whether these forms, together with additional cases, generalize into design knowledge for computational fabrication tool design.

In parallel with the remaining evaluation of *Cross-Section Weaving*, I am developing a third system that extends the *creativity-expansive tooling* thesis to a new material process and computational approach—complementing woodworking in *Uniquely Shaped Spaces* and weaving in *Cross-Section Weaving*. This system serves as a deliberate stress test of the generative and representational orientations identified in the first two cases. *Uniquely Shaped Spaces* foregrounds generative engagement but is also representational—participants see and interpret illustrated shelving configurations. *Cross-Section Weaving* foregrounds representational engagement but is also generative—a simple line drawing compiles into a complete weaving draft. The third system takes a different approach to these same orientations—one where the boundary between generating and interpreting is less clearly staged—probing the limits and generality of the *creativity-expansive tooling* concept rather than simply adding a third instance of it. More broadly, this work contributes to ongoing conversations in design research about how computational tools shape creative agency—extending to other contexts where practitioners negotiate with algorithmic behavior and material response.

5 Discussion

I envision fabrication tools where the practitioner’s encounter with computational behavior and material response is the creative process—not an obstacle to be streamlined, but the site where new possibilities emerge. Realizing this vision surfaces several live tensions I hope to discuss. My analytical unit—*moments of negotiation*—is designed to be legible across technical and qualitative registers, but the forms of evidence it relies on (process traces, exemplar artifacts, fabrication logs) speak more naturally to design research than to technical HCI; I am interested in how to present this analysis compellingly across venues without flattening what makes each form of evidence meaningful. A related tension concerns the cross-case comparison: the two systems foreground different orientations of computational engagement—generative in *Uniquely Shaped Spaces*, representational in *Cross-Section Weaving*—but neither is purely one or the other. How should the analysis handle forms of engagement that overlap across systems, and what does it mean for the *creativity-expansive tooling* concept if these orientations are matters of emphasis rather than kind? The study design for *Cross-Section Weaving* makes this tension concrete: what counts as evidence of negotiation when the tool’s engagement is representational—changing how a practitioner thinks—rather than generative? Finally, the interlocutor framing raises a question about its relationship to existing accounts of negotiation in making: Andersen et al.’s digital crafts-machine-ship [1] and Devendorf and Ryokai’s being the machine [7] both foreground shared agency between practitioner and computational process. What does designing tools as interlocutors add beyond these accounts, and where does the distinction matter?

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